BIO301 FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS (2017-2022)

GROUP: BIOTECH BRAINY BUNCH

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SUBJECTIVE CURRENT 2022

1. **Define the terms Autosomal dominant and autosomal recessive?**

Autosomal Dominant: Affected individuals are heterozygote. One allele is mutated, while other one is normal.

Autosomal recessive: Homozygous, when both alleles of the gene are mutated.

2. **X-linked recessive and dominant?**

X-linked recessive: Mutated alleles are present on either of the sex chromosomes.

X-linked dominant: inheritance refers to genetic conditions associated with mutations in genes on the X chromosome.

Families with an X-linked recessive disorder often have affected males, but rarely affected females, in each generation. For X-linked dominant diseases, however, a mutation in one copy of an X-linked gene will result in disease for both males and females.

3. **Principle of PCR and its reagents name? (5)**

Principle of PCR

The PCR technique is based on the enzymatic replication of DNA. In PCR, a short segment of DNA is amplified using primer mediated enzymes. DNA Polymerase synthesises new strands of DNA complementary to the template DNA. The DNA polymerase can add a nucleotide to the pre-existing 3'-OH group only.

The **key ingredients** of a PCR reaction are Taq polymerase, primers, template DNA, and nucleotides (DNA building blocks). The ingredients are assembled in a tube, along with cofactors needed by the enzyme, and are put through repeated cycles of heating and cooling that allow DNA to be synthesized.

4. Function of tRNA?

Transfer RNA (tRNA) has a long-established role in protein synthesis. The tRNA molecule serves as an adaptor between the genetic instructions written in nucleic acid sequences and the protein products encoded in genes.

5. What is point mutation?

A **point mutation** or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

6. Steps of organic method?

PCR is performed on thermocycler and it involves three main steps:

(1) Denaturation of dsDNA template at 92–95°C,

(2) Annealing of primers at 50–70°C, and

(3) Extension of dsDNA molecules at approx. 72°C. These steps are repeated for 30–40 cycles.

7. Diff b/w hot start and suicide pcr?

Hot start technique is based on the principle of initiating the polymerase reaction at or above the primer annealing temperature, thereby preempting possible extensions of the primers nonspecifically annealed at ambient temperature (20°-55°C) at which Taq Pol retains partial activity. For some applications, hot-start PCR may be the only way to amplify successfully the desired product(s) with high yield and specificity.

Suicide PCR method prescribes the use of any primer combination only once in a PCR (hence the term "suicide"), which should never have been used in any positive control PCR reaction, and the primers should always target a genomic region never amplified before in the lab using this or any other set of primers.

8. Co-dominance and three examples?

Codominance, as it relates to genetics, refers to a type of inheritance in which two versions (alleles) of the same gene are expressed separately to yield different traits in an individual.

1) AB Blood Type.
2) Horse color.
3) Flower colors.

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9. Nucleotides numbers, initiation codon, stop codon?

Nucleotide Numbering: The nucleotides are shown with standard numbering convention. The aromatic base atoms are numbered 1 through 9 for purines and 1 through 6 for pyrimidines. The ribose sugar is numbered 1' through 5'.

The **start codon** is the first codon of a messenger RNA transcript translated by a ribosome. The start codon always codes for methionine in eukaryotes and Archaea and a N-formylmethionine in bacteria, mitochondria and plastids. The most common start codon is AUG

There are 3 **STOP codons** in the genetic code - UAG, UAA, and UGA. These codons signal the end of the polypeptide chain during translation. These codons are also known as nonsense codons or termination codons as they do not code for an amino acid.

10. Three mitochondrial diseases?

- Myoclonic epilepsy
- Mitochondrial recessive ataxia
- Leber hereditary optic neuropathy
- Sensory ataxia neuropathy

11. Function of STRP?

STR is simple repeats of nucleotide sequences.

1) Mononucleotide, dinucleotide
2) Used in linkage analysis
3) Used in DNA forensics

12. Name of genes that are used for gene mapping of drosophila?

Find the answer

13. Frame shift mutation?

A frameshift mutation is a genetic mutation caused by indels of a number of nucleotides in a DNA sequence that is not divisible by three. Due to the triplet nature of gene expression by codons, the insertion or deletion can change the reading frame, resulting in a completely different translation from the original.

14. Enlist number of chromosomes in any 10 species?

1) Fruit fly (Drosophila melanogaster) 8
2) Nematode(Caenorhabditis elegans) 12/11
3) Pea (Pisum sativum) 14
4) Aloe vera 14
5) Kangaroo 16
6) Carrot (Daucus carota) 18
7) Snail 24
8) Rice (Oryza sativa) 24

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9) Tomato (Solanum lycopersicum) 24
10) Sunflower (Helianthus annuus) 34

15. Hot start pcr in detail? (5)

Hot Start PCR is a technique that inhibits Hot Start Taq polymerase activity or the incorporation of modified dNTPs during reaction set up until a heat activation step occurs. Hot Start PCR allows for reaction set up at room temperature without non-specific amplification and primer dimer formation.

HOT START PCR PROTOCOL

a) Antibody-Mediated Hot Start PCR Protocol
b) dNTP-Mediated Hot Start PCR Protocol

The basic principle of Hot Start PCR methods and reagents is that they are designed to inhibit Hot Start Taq DNA polymerase activity, or the incorporation modified dNTPs during reaction set up until a heat activation step occurs. Various methods are available to halt Hot Start polymerase activity and include chemical modification, antibody-mediated, and aptamer-mediated technology.

16. What is addition mutation?

Additions mutation: Addition mutation is actually results in a frame shift mutation. Often, it has severe effects on protein function. It is also called as insertions.

17. Define homozygous condition?

Homozygous, as related to genetics, refers to having inherited the same versions (alleles) of a genomic marker from each biological parent. Thus, an individual who is homozygous for a genomic marker has two identical versions of that marker.

18. What is DNA loop in E.coli?

A DNA loop occurs when a protein or a complex of proteins simultaneously binds to two different sites on DNA with looping out of the intervening DNA

19. Describe Mendel's work heredity and variations?

Gregor Mendel, through his work on pea plants, discovered the fundamental laws of inheritance. He deduced that genes come in pairs and are inherited as distinct units, one from each parent. Mendel tracked the segregation of parental genes and their appearance in the offspring as dominant or recessive traits.

20. What is loss of function and gain of function?

Loss of function: Gain of function Loss of function: Complete or partial loss of the normal function. Structure of protein is so altered that it is no longer works correctly. Mutation can occur

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in regulatory regions that affect transcription, translation or splicing of the protein. Frequently recessive

Gain of function: Produces an entirely new trait. Such a mutation causes a trait to appear in inappropriate tissues or at inappropriate times in development. It is frequently dominant.

21. What are genetic disorders?

A genetic disorder is a genetic problem caused by one or more abnormalities in the genome. Most genetic disorders are quite rare and affect one person in every several thousands or millions. Genetic disorders may be hereditary, meaning that they are passed down from the parents' genes.

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1. **Function of Pedigree?**

A pedigree may be used to establish the probability of a child having a particular disorder or condition. It may be used to discover where the genes in question are located (x, y, or autosome chromosome), and to determine whether a trait is dominant or recessive.

2. **Define Epigenetics?**

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence

3. **RAPD and ALFP Difference?**

RAPD is Random Amplification of Polymorphic DNA'. It is a type of PCR, but the segments of DNA that are amplified are random.

AFLP is a PCR-based tool used in genetics research, DNA fingerprinting, and in the practice of genetic engineering. Developed in the early 1990s by Key gene, AFLP uses restriction enzymes to digest genomic DNA, followed by ligation of adaptors to the sticky ends of the restriction fragments.

4. **Complications of Inheritance?**

Complications to inheritance patterns: Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance. Complications: A common recessive inheritance can give dominant pattern, Autosomal dominant inheritance with non-penetrance, Autosomal dominant inheritance with variable expression.

5. **Mutation Effects in Functioning Of Genes?**

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Some mutations don't have any noticeable effect on the phenotype of an organism. This can happen in many situations: perhaps the mutation occurs in a stretch of DNA with no function, or perhaps the mutation occurs in a protein-coding region, but ends up not affecting the amino acid sequence of the protein.

6. Causes of Epigenetics?

Stress Hormone Causes Epigenetic Changes. Researchers found that chronic exposure to a stress hormone causes modifications to DNA in the brains of mice, prompting changes in gene expression

7. Genetic Mapping In Humans?

Genetic mapping - also called linkage mapping - can offer firm evidence that a disease transmitted from parent to child is linked to one or more genes. Mapping also provides clues about which chromosome contains the gene and precisely where the gene lies on that chromosome.

8. Point Mutations and Enlist 3 Structural Changes In Chromosomes? (REPEAT)

A point mutation or substitution is a genetic mutation where a single nucleotide base is changed, inserted or deleted from a sequence of DNA or RNA. Point mutations have a variety of effects on the downstream protein product—consequences that are moderately predictable based upon the specifics of the mutation.

Deletion, duplication, translocation, transcription.

9. Modification Of Epigenetics

Epigenetic modification - non coding RNA is a new mechanism for gene regulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs). Three common types of modification - Cytosine methylation, Histones modifications, siRNAs

10. Mutation Alter Genes Expression?

Mutation and chromosome rearrangement can produce changes in DNA sequence which has been identified in some cancer cells. Recombination and non-disjunction can lead to the expression of recessive alleles. Overexpression of genes by amplification has also been observed in certain tumors.

11. Sickle cell anemia?

Sickle cell anemia (sickle cell disease) is a disorder of the blood caused by inherited abnormal hemoglobin (the oxygen-carrying protein within the red blood cells). The abnormal hemoglobin causes distorted (sickled) red blood cells. The sickled red blood cells are fragile and prone to rupture.

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Cause:

The protein may lose its function, which can result in a disease in the organism. For example, sickle-cell disease is caused by a single point mutation (a missense mutation) in the betahemoglobin gene that converts a GAG codon into GUG, which encodes the amino acid valine rather than glutamic acid.

12. Mutation nomenclature?

Standard nomenclature based on genomic DNA reference sequences and RNA reference sequences is not shown. "Genomic DNA reference sequence" simply indicates any human DNA sequence in the database that is not based on a cDNA sequence. Example of standard mutation nomenclature based on a coding DNA sequence.

13. VTRN and application?

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes, Show variations in length between individuals .Applications of VNTR Microbiology DNA fingerprinting, Genetic diversity, Forensics Mapping of genomes, Breeding patterns of wild or domesticated animals.

14. Define Epigenetic? 2 marks

Epigenetics literally means 'above' the genetics. Epigenetic trait is a stable and heritable phenotype resulting from changes in chromosome without alterations in the DNA sequence.

15. Complications of inheritance? 3 marks

- Multiple genes - epistasis, polygenic traits
- Genes & the environment – sex influenced traits, incomplete penetrance
- A common recessive inheritance can give dominant pattern
- Autosomal dominant inheritance with non-penetrance
- Autosomal dominant inheritance with variable expression.

16. How mutations alter gene function? 10 marks

Splice site mutations have severe effects

Splice sites site mutations usually have severe effects on protein functions. Often this kind of mutation produces one or more exon skipping or larger fragments of DNA.

Effects of null allele and hypo morph

Null allele results in loss of function. Usually no gene product hypo morph alleles results in decreased activity of the proteins.

Effects of mutation on gene function

Gain of function - increased activity. Dominant negative antagonizes normal product. Neomorph novel activity.

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Missense mutations have sometimes severe effects on protein function while some times less severe effects. It depends on the type of amino acid which was replaced.

17. Epigenetic modifications (10)

Cytosine methylation

Histones modification

Non-coding RNAs (siRNAs)

All above regulate gene expression.

Cytosine methylation occurs at upstream regions of genes

Often located just upstream of genes (promoter regions). Associated with attenuation of expression of nearby genes.

Histones modifications

Histones are the proteins that organize the genetic material. Have a high percentage of basic amino acids, which gives histones an overall positive charge, while DNA is negative charged.

Acetylated histones-genes are active

When histones are acetylated, chromatin is open and genes are potentially active. When histones not tagged, deacetylated, the chromatin condenses and genes silenced.

Histones modifications:

Most histones modifications occur on the extended tails of histone proteins. HISTONES **acetylation**

Two enzyme types are involved in histone acetylation. HAT: Histone acetyltransferase HDAC: Histone deacetylase.

Epigenetic modification - non coding rna

It is new mechanism for gene regulation. RNA which is not used for making proteins can be cleaved and used to inhibit protein-coding RNAs (siRNAs, microRNAs).

18. Chromosomal aberration 10 marks?

ABERRATIONS: Any change in normal number or structure of chromosomes. It affects individuals by altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes.

CLASSIFICATION:

Numerical (It is usually due to De-novo error in meiosis).

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Structural (may be due to de novo error in meiosis or inherited). Different cell lines (occurs postzygotically).

NUMERICAL CHANGES

- **Aneuploidy**
  - Hypoploidy
  - Hyperploidy
- **Euploidy**
  - Monoploidy
  - Diploidy
  - Polyploidy

STRUCTURAL

Translocations Deletions Duplications Inversions

19. Monosomy

Monosomy is the state of having a single copy of a chromosome pair instead of the usual two copies found in diploid cells

Causes of monosomy: followings are the causes of monosomy

Non disjunction – one gamete receives two copies of homologous chromosomes and other will have no copy. Loss of chromosome - As it moves towards pole of cell during anaphase. Fruit fly **monosomy**, is the cause of some human diseases such as Turner syndrome and Cri du Chat syndrome

20. What are structural abnormalities?

Structural abnormalities are when part of an individual chromosome is missing, extra, switched to another chromosome, or turned upside down.

Types based on genetic material

Balanced – chromosome complement is complete.

Unbalanced – when there is incorrect amount of genetic material.

Types: Translocations, Deletions, Insertions, Inversions, Ring chromosome, Isochromosome

21. Parents of affected individuals are asymptomatic

In most of the cases, parents of affected individuals are asymptomatic deletions. For point deletions, one nucleotide has been deleted from the sequence. Cystic fibrosis, Turner syndrome, Results in frame shift mutation.

22. Suppressor mutation

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Suppresses the effect of other mutation occurs at a site different from the site of original mutation.

23. Intragenic suppression

Intragenic suppression results from suppressor mutations that occur in the same gene as the original mutation. Intergenic (extragenic) suppression relieves the effects of a mutation in one gene by a mutation somewhere else within the genome. Second mutation is not on same gene as original mutation.

24. Penetrance and Expressivity

Penetrance measures the proportion of individuals in a population who carry a specific gene and express the related trait. Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.

In genetics, expressivity is the degree to which a phenotype is expressed by individuals having a particular genotype. (Alternately, it may refer to the expression of particular gene by individuals having a certain phenotype.)

25. Low and full Penetrant

Low Penetrance An allele with low penetrance will only sometimes produce the symptom or trait with which it has been associated at a detectable level. In low penetrance, it is difficult to distinguish environmental from genetic factors.

Highly Penetrant If an allele is highly penetrant, and then the trait it produces will almost always be apparent in an individual carrying the allele.

26. Polyploidy

Polyploidy is the heritable condition of possessing more than two complete sets of chromosomes. Multiples of haploid cells or no of chromosomes are called polyploidy. It can be in form of triploidy, tetraploidy , pentaploidy etc. during polyploidy fetus does not survive this phenomena is seems in many crops.

Two major types

- Autopolyploidy
- Allopolyploidy

27. Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

28. Frequency of autosomal recessive disorders

Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

29. Trisomy

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Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two. Trisomy is a type of aneuploidy. Trisomy of chromosome 21, which is found in Down syndrome, is called trisomy 21

30. Inversion mutation

Inversion mutations only affects a small part of the gene. May a single nucleotide or more than one nucleotides are involved.

31. Types of X-linked recessive disorders

- Duchene Muscular Dystrophy
- Hunter's Disease
- Menkes Disease
- Hemophilia A and B
- Color Blindness

32. Diagnostic tests for genetic disorders

Physical analysis: Physical analysis for affected individual

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus.,

33. Polymorphism

Polymorphism, in biology, a discontinuous genetic variation resulting in the occurrence of several different forms or types of individuals among the members of a single species

Aneuploidy: Change in chromosomes number can occur by the addition/deletion of chromosome or part of a chromosome

Euploidy: Gain of one or more complete sets of Chromosomes

34. Pattern of inheritance

Most genes have one or more versions due to mutations or polymorphisms referred to as alleles. There are five basic modes of inheritance for single-gene diseases: autosomal dominant, autosomal recessive, X-linked dominant, X-linked recessive, and mitochondrial.

35. Multifactorial diseases six

Genetic disorders controlled by many genes plus the effects of the environment.

Asthma, Cancers ,Diabetes ,Heart diseases, Hypertension, Obesity

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36. Pleiotropy

The ability of a gene to affect an organism in many ways is called pleiotropy.

Pleiotropy refers to an allele which has more than one effect on the phenotype.

37. Types of deletion include the following:

- Terminal deletion – a deletion that occurs towards the end of a chromosome.
- Intercalary/interstitial deletion – a deletion that occurs from the interior of a chromosome.
- Micro deletion – a relatively small amount of deletion (up to 5Mb that could include a dozen genes).

38. Applications of molecular markers – short range

- Parentage determination
- Genetic distance estimation
- Determination of twin zygosity

39. Applications of Molecular Markers – Long Range

- Gene mapping & mapping of QTL by linkage
- Markers assisted selection of breeds

40. Method of mapping gene disorders

Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

41. How genetic mapping performed

Scientists isolate DNA from samples and examine it for the unique patterns of bases. Molecular techniques are referred to as genotyping polymorphisms or markers.

42. Single and Multifactorial Mutation?

Single gene disorders are caused by DNA changes in one particular gene, and often have predictable inheritance patterns. Over 10,000 human disorders are caused by a change, known as a mutation in a single gene. These are known as **single gene** disorders

Multifactorial Disorders. Researchers are learning that nearly all conditions and diseases have a genetic component. Some disorders, such as sickle cell disease and cystic fibrosis, are caused by mutations in a single gene. Conditions caused by many contributing factors are called complex or multi factorial disorders

43. Autosomal Dominant Disordered?

A single abnormal gene on one of the first 22 non-sex (autosomal) chromosomes from either parent can cause an autosomal disorder. **Dominant inheritance** means an abnormal gene from

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one parent can cause disease. Examples of autosomal dominant disorders include Marfan syndrome and neurofibromatosis type 1.

44. During DNA Extraction How Proteins Are Denatured?

Adding proteinase

45. Define Locus?

A locus (plural loci) in genetics is a fixed position on a chromosome, like the position of a gene or a marker (genetic marker). Each chromosome carries many genes; human's estimated 'haploid' protein coding genes are 19,000–20,000, on the 23 different chromosomes.

46. Sex Linked Traits And Disorder?

Genes on the X or Y chromosome are called sex-linked. ... For example, an X-linked recessive allele in humans causes haemophilia, which is much more common in males than females because males are hemizygous and therefore express the trait when they inherit one mutant allele.

47. Define Multiple Alleles With Example?

Multiple alleles three or more alternative forms of a gene (alleles) that can occupy the same locus. However, only two of the alleles can be present in a single organism. For example, the ABO system of blood groups is controlled by three alleles, only two of which are present in an individual.

48. What Is Location Of Intronic Sequence With Respect To Exons?

Sequences that are joined together in the final mature RNA after RNA splicing areexons. Introns are found in the genes of most organisms and many viruses, and can be located in a wide range of genes, including those that generate proteins, ribosomal RNA (rRNA), and transfer RNA (tRNA).

49. Is There Any Rules For Mutation Nomenclature?

Standard mutation nomenclature based on a “genomic DNA reference sequence” requires a prefix “g.” and numbering starts with number 1 for the first nucleotide in the file. Example of nucleotide numbering based on a coding DNA sequence.

50. Structural change in chromosome?

Structural Change in the Structure of Chromosomes: Chromosomes are the vehicle of hereditary material or genes. Any alteration, addition or deletion of chromosomal part leads to alteration of number, position or sequence of genes in the chromosome.

51. Define linkage?

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Linkage is defined genetically as the failure of two genes to assort independently. Linkage occurs when two genes are close to each other on the same chromosome.

52. What is polygenic inheritance?

Condition when two or more genes influence the expression of one trait. Polygenic inheritance is controlled by multiple genes

53. What is epistasis?

Epistasis is a phenomenon that consists of the effect of one gene being dependent on the presence of one or more modifier genes Type of polygenic inheritance. Where alleles at one gene locus can hide or prevent expression of alleles at second gene locus.

54. What is linked gene?

Genes that are found on the same chromosomes are called linked gene.

55. What is physical mapping?

A physical map is primarily base on the location of landmark along a DNA molecule.

56. write Name of different mapping technique?

1: Cytogenetic map 2: In situ hybridization 3: Genetic mapping 4: Restriction enzyme 5: Radiation hybrid mapping

57. What is position independent?

It is a forward genetics. It started at the phenotype, determining which protein was involved and getting to the gene through the protein.

58. How gene - species oligonucleotide are prepared?

Oligonucliotide isolating the protein product of the gene. These genes are usually produced large amount of protein. Then gene - species oligonucleotide are prepared.

59. How polygenic inheritance are controlled by multiple genes?

When two or more genes influence the expression of one trait. Then polygenic inheritance is controlled by multiple genes.

60. What is multifactorial?

Many of traits are effected by the person,s environment, so they are called multifactorial.

61. Give example of polygenic trait?

1: Stature 2: Shape of face 3: Finger prints

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62. How many genes are controlled the chicken combs shapes? And where these found?

Comb shapes are controlled by two genes. These genes are found on two different chromosomes. Gene

- 1: R, Gene
- 2:p.

63. Short note on hair colour?

Hair color is controlled by alleles on chromosomes 3, 6, 10 and 18. When more dominant alleles appear in genotype, the darker will be the hair.

64. What is role of positive epistasis?

Positive epistasis is occurring between beneficial mutations. It generates improvement in function.

65. What is negative effect?

When two mutation has a smaller effect than expected from their effects when alone. It is called negative effect.

66. How sign epistasis occur?

It occur when one mutation has the opposite effect when in the presence of another mutation.

67. What is dominant epistasis?

It is fruit color in summer squash. Epistasis gene exerts is effect with the presence of a dominant allele.

68. Write duplicate dominant epistasis and its shape?

Fruit shape in shephered,s purse. A-or B-=heart shape. aa and bb = narrow shape

69. Write molecular mechanism of epigentics?

1: Methylation of cytosine posttranslational modification to histone protein , include acetylation, phosphorylation .

70. How many epigenetic modifications are there write their name?

Three types of epigenetic modifications are there these are

1) Cytosine methylation

2) Histones modification

3)Non-coding RNAs (siRNAs)

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71. Describes histones in Histone modification?

Histones are the proteins that organize the genetic material. Have a high percentage of basic amino acids, which gives histones an overall positive charge, while DNA is negative charged.

72. Which two enzymes involved in Histone accyitylation?

1) HAT: Histone acetyltransferase

2) HDAC: Histone deacetylase.

73. What is role of histones in accetylation?

When histones are acetylated, chromatin is open and genes are potentially active. When histones not tagged, deacetylated, the chromatin condenses and genes silenced.

74. What are common Epigenetics regulated phenomena?

1) Cellular Differentiation

2) X Chromosome inactivation 3) Genomic imprinting

75. Describe Identical twins?

Identical twins are from the same zygote, so they begin life with the same genetic information. While infants they experience the same or very similar environments, so there is little variation in the epigenome. Over time the twins' environments will diverge, resulting in individual epigenetic tags to form for each twin. The difference in the twins' epigenomes is what makes them become different when they are older. The epigenetic tags can have such an effect on the twins that one can develop a disease while the other is fine.

76. What is Vorinostat?

Vorinostat is a histone deacetylase inhibitor.

77. What environmental factors affected by epigenetics?

Nutrition of mother, stress mother, social interactions and physical activities, Exposure to toxins and diet.

78. Defines penetrance?

Proportion of individuals carrying a particular variant of a gene (allele or genotype) that also expresses an associated trait (phenotype)

79. What is variable expressivity?

When phenotype is expressed to a different degree among individuals with the same genotype it is called variable expressivity.

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80. MEN 1 stand for?

Multiple endocrine neoplasia 1 it is a hereditary disorder

81. Write any two types of penetrance?

Complete penetrance, highly penetrance, incomplete or reduced penetrance, low penetrance.

82. What is chromosomal aberration?

Any change in normal number or structure of chromosomes. Affects individuals may be due altering the amounts of products of the genes involved. Altered amounts may cause anomalies directly or may alter the balance of genes

83. How many types of chromosomal aberration are there describe them?

1) Numerical (usually due to de novo error in meiosis).

2) tructural (may be due to de novo error in meiosis or inherited).

84. Which structural abnormalities may occurs due to change in chromosomes?

Structural abnormalities may occurs due to

1) Translocations

2) Deletions

3) Duplications

4) Inversions

85. Chromosomal identifications dependent upon which factors? Write their name?

1. Size

2. Location of the centromere

3. Banding patterns

86. How many chromosomes are present in down syndrome?

47 chromosomes

87. Write name of Common numerical Autosomal and sex chromosome Abnormalities?

Autosomal 1) Downsyndrome (Trisomy 21) 2) Edwards syndrome (trisomy 18) 3) patau syndrome (trisomy13) sex chromosomes abnormalities 1) Turner syndrome 45,x 2) klinefelter syndrome 47,xxy

88. Write the characteristics of Down syndrome

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it is a trisomy of chromosomes no 21. The frequency of trisomy increases with increasing maternal age. Sometimes, robertssonian translocation involving chromosome 21. 3-4%, not related to maternal age.

89. Write characteristic of turner syndrome.

Relatively common disorder caused y the loss of genetic material from one of the sex chromosomes. Affects only females and they have only one x chromosome.

90. Define Triploidy ? how many chromosomes that deseas contain?

This is phenomena in which all the set of chromosomes contain one extra chromosome like a normal person have 2 pair of chromosomes and triploidy person have 3 chromosomes so they become 69 chromosomes and a normal person has 46 chromosomes.

91. Define structural abnormalities?

Rearrangement because of chromosome breakage and subsequent reunion in a different configuration.

92. What are the types of structural abnormalities?

There are 6 types i.e 1) Translocation 2) Deletions 3) insertions 4) inversions 5) ring chromosomes 6) iso chromosomes

93. Define Translocation? Write its name?

Transfer of genetic material from one chromosome to another chromosome called translocation. There are two types of translocations 1) Reciprocal translocation and 2) Robertosnian Translocation.

94. Defien Reciprocal Translocation?

An exchange of genetic material between two different chromosomes is called a reciprocal translocation in which no loss or gain of chromosomal material.

95. Define Robertsonian translocations.

Break occurs at the extreme end of the short arms of two non homologuous acrocentric chromosomes and the small acentric fragments are lost so the larger fragments fuse at their centromeric regions to for a single chromosome. This type of translocation occurs between chromosomes no 13 and 14 in humans.

96. Writes common human diesease caused by translocations?

The common diseases in humans caused by translocations are

1) Cancer 2) infertility 3) down syndrome 4) leukemia 5) XX male syndrome

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97. What is difference between disjunction and non-disjunction?

Separation of two homologous chromosomes during cell division is called disjunction. When two homologous chromosomes or sister chromatids fails to separate during cell division it is called as non-disjunction.

98. Differentiate between meiotic and mitotic non disjuntion?

If non-disjunction occurs during 1st meiotic division then all the sperms/ova derived from primary spermatocyte loocytes will be abnormal.

Mitotic non-disjunction results in cells with different number of chromosomes due to the breaking of the spindle fibers during metaphase or anaphase

99. Explain the failure of chromosome separation?

1) Failure of a pair of homologous chromosomes to separate in meisis1.

2) Failure of sister chromatids to separate during meiosis II.

3) Failure of sister chromatids to separate during mitosis.

100. What will be the results in Mitotic non-disjunction?

Mitotic non-disjunction results in trisomy and monosomy. it causes trisomic and monosomic daughter cells to be formed.

101. What are the causes of Non-Disjunciton ?

There are several cause of non-disjunction but the most are 1) the Aging effect on primary oocyte 2) radiation from the environment 3) delayed fertilization after ovulation.

102. What is the difference between Trisomy and Monosomy?

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two. Trisomy is a type of aneuploidy (an abnormal number of chromosomes).

Monosomy occurs when there is a loss of a single chromosome. monosomy of autosomes is lethal during sex chromosome monosomy will be Turner syndrome XO in which loss of sex chromosomes and they don't survive and died

103. Define Trisomy Types

There are 4 types of Trisomy.

1) Primary Trisomy - An entire etra chromosome has been copied it is called primary trisomy.

2) Partial trisomy, in which an extra copy of part or fragment of a chromosomes present called partial trisomy.

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3) Secondary trisomy- the extra chromosome has duplicated arms so it is called secondary trisomy.

4) Tertiary trisomy - extra chromosome is made up of copies of arms from two other chromosomes

104. What is the common Trisomy in humans?

Trisomy 21, Trisomy 18 trisomy 13 and Klinefelter's syndrome (47xxy)

105. What are the causes of Monosomy?

during Non-disjunction one gamete receives two copies of homologous chromosomes and other will have no copy. loss of chromosome as it move towards pole of cell during anaphase.

106. Write the name of Monosomy diseases in humans.

Turner syndrome and Cri Du chat syndrome

107. Write types of polyploidy in plants.

Triploid (3x) Watermelon, Tetraploid(4x) cotton, Pentaploid (5x) Kenai Birch , Hexaploid (6x) Wheat

108. Write the causes of Polyploidy

1) Retention of polar body. 2) Formation of diploid sperms 3) Di-spermy- fertilization by two sperms.

109. Write polyploidy examples in plants.

Triploid - Apple , Bannana, citrus fruits, ginger watermelons.

Tetraploid. Cotton, patato, tobacco, peanut

Hexaploid - Oat, Kiwifruit

Octaploid- Straberry

Decaploid-Some sugar cane hybrids

110. What is difference between Autopolyploidy and Allopolyploidy.

Autopolyploids are polyploids with multiple chromosome sets dervied from a single species. in case of allopolyploids are polypoloids with chromosomes derived from different species in the result of cross between two different species. Interspecies crosses can generate alloploids and their offspring generally sterile.

111. Which chemical is used to promote the polyploidy in plants?

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Colchicine is used to promote polyploidy in the plants to avoid non disjunction.

112. What does the sterile means?

The word sterile means that the specie cannot reproduce.

113. What is the difference between Aneuploidy and Euploidy?

Aneuploidy:- change in chromosome number can occur by the addition /deletion of a chromosome or part of a chromosome. In thw case of Euploidy :- a gain of one or more complete sets of chromosomes

114. What is the difference between Terminal and interstitial deletions?

In the Terminal, the part of the chromosomes that is at the end like tips of the chromosomes been deleted.

In the interstitial, the part of chromosomes which is at the middle part or two breaks.

115. Define Gene or chromosomal duplication?

Gene duplication or chromosomal duplication is a major mechanism through which new genetic material is generated during molecular evolution

116. Write the causes of chromosomal duplications?

1) DNA Replication - Gene duplications can arise as products of several types of errors in DNA replication and repair machinery.

2) Ectopic homologous recombination;- common sources of gene duplications include ectopic homologous recombination

3) Ectopic recombination Misaligned homologous - Duplication arises from an event termed unequal crossing over that occurs during meiosis between misaligned homologous chromosomes.

4) Other causes of duplications - Retro-transposition event, aneuploidy, polyploidy, and replication slippage are also cause of duplications

117. Define Dispersed and Tandem Duplications.

Some duplication is dispersed found in different locations from each other. Tandem duplications found next to each other

118. What is called pseudogenes?

Pseudo genes are inactive forms of gene forms by mutations, these genes cannot produce messenger RNA.

119. What is called Chromosomal inversions?

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It involves two break re-arrangement and segment is reversed there are two types Pericentric inversions in which centromere is involved and paracentric in which only chromosomes Arms like Por Q is involved.

120. What are the chromosomal inversions affects?

It affects gene expression gene is re-positioned in a way that alters its gene expression 2 % of the human population carries karyotipically detectable inversions.

121. Define Heterozygote inversion?

Individual with one copy of a normal chromosome and one copy of an inverted chromosome called inversion heterozygotes usually phenotypically they are normal.

122. What are the effects of chromosomal inversion in humans?

The most common inversion seen in humans is on chromosome 9, at inv 9 (p12q13) this inversion is generally considered to have no harmful effects but increased risk of miscarriage or infertility.

123. Mutations are source of genetic variation? Justify

Mutation and variation Mutations are a major source of genetic variation in a population increasing biodiversity. Some variations may help to survive better and longer. Mutations in gametes (egg & sperm) are passed onto offspring. Mutations in body cells only affect the organism in which they occur and are not passed onto offspring.

124. Write name of major types of mutations.

Major types of mutations

1) Points Mutations 2) Variations in chromosomes number 3) Variations in chromosomes structure

125. writes differences between Somatic and Germline mutation

Somatic mutation: Arise in the somatic cells. Passed on to other cells through the process of mitosis. Effect of these mutations depends on the type of the cell in which they occur.

Germline mutation: They occur in the cells that produce gametes. This type of mutation usually passed on to future generation. In multicellular organisms, the term mutation is generally used for germ line mutations..

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Individual with one copy of a normal chromosome and one copy of an inverted chromosome called inversion heterozygotes usually phenotypically they are normal.

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128. Differentiate between ring chromosome and Philadelphia chromosome?

A ring chromosome is a chromosome whose arms have fused together to form a ring. in humans it is denoted by small r and in drosophila it is denoted by capital R.

Philadelphia chromosome:- is a specific chromosomal abnormality that is related with chronic myelogenous leukemia(CML). Reciprocal translocation between chromosome 9 and chromosome 22 which is specifically designated t(9:22)(q34:q11)

129. Write name of major types of mutations.

Major types of mutations

1) Points Mutations 2) Variations in chromosomes number 3) Variations in chromosomes structure

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Somatic mutation: Arise in the somatic cells. Passed on to other cells through the process of mitosis. Effect of these mutations depends on the type of the cell in which they occur.

Germline mutation: They occur in the cells that produce gametes. This type of mutation usually passed on to future generation. In multicellular organisms, the term mutation is generally used for germ line mutations..

131. Explain - Forward, Reverse and Neutral mutations

Wild type most common phenotype in natural populations of the organism is called as wild type phenotype. The effect of mutation is considered with reference to wild type phenotype.

Forward mutation: Mutation that alters the wild type phenotype into mutant phenotype.

Reverse mutation: Mutation that changes a mutant phenotype back into the wild type.

Neutral mutation: Mutation that alters the amino acid sequence of the protein but does not change its function as laced amino acid is chemically similar or the affected AA has little influence on protein function..

132. Differentiate between conditional and lethal mutants?

Conditional mutation: Expressed only under certain conditions. E.g. A temperature-sensitive mutation can cause cell death at high temperature, but might have no deleterious effects at a lower temperature.

Lethal mutation: Causes the death of the organism. Death does not have to occur immediately, it may take several months or even years. Longevity of an individual is significantly reduced.

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133. Write difference between spontaneous and induced mutations?

Spontaneous mutation: Mutations that result from natural changes in DNA.

Induced mutation results from changes caused by environmental, chemicals and radiations.

134. Define mutagens?

Mutagen: Any environmental agent that increases the rate of mutation above the spontaneous is called a mutagen such as chemicals & radiations, X-rays, Mustard gas etc.

135. Define Base analogs?

Base analogs: Chemicals with structures similar to that of any of the four standard bases of DNA. DNA polymerases cannot distinguish these analogs. They may be incorporated into newly synthesized DNA molecules.

136. What is difference between Transition and transversion mutation.

If purine (A/G) is replaced by purines and pyrimidine (T/C) is replaced by pyrimidines it is called transitions mutation. If purine substitutes for pyrimidine or vice versa is called transversion.

137. Define Silent Mutation

Silent Mutation: These types of mutations have affected the DNA sequence but not the protein. They have no effect on the organism's phenotype. There is change at DNA level but no change in protein level.

138. Define Missense mutation and its effects on protein formation?

Missense mutation: Missense mutations substitute one amino acid for another. Some missense mutations have very large effects. Some have minimal or no effect. It depends on where the mutation occurs in the protein's structure and how big a change in the type of amino acid it is. Missense mutation produces a change in amino acid sequence in protein product (e.g. Histidine in for Arginine). May change function of protein or may not!

139. Define nonsense mutation and its effects on protein formation

Nonsense mutation: Nonsense mutations convert an amino acid into a stop codon. The effect is to shorten the resulting protein. Sometimes this has only a little effect, as the ends of proteins are often relatively un-important to function. However, often nonsense mutations result in completely non-functional proteins.

140. Define Deletion Mutations and its effects.

Deletions mutations:- Deletion of one or more nucleotides in DNA sequence is called deletion mutations. These types of mutations have severe effects on the phenotype.

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141. How mutations alter gene function

Null allele (loss of function) - no gene product.

Hypomorph - decreased activity.

Gain of function - increased activity.

Dominant negative-antagonizes normal product

Neomorph - novel activity

Predicting that a gene product won't do the job

Deletion, nonsense, frameshift of sequence is deleterious.

Mutation in splice site usually has sever effects.

Missense mutations depend on location in protein.

142. Explain the Mutation nomenclature

All variants should be described at the most basic level, i.e. DNA level.

Descriptions should always be in relation to a reference sequence, either a genomic or a coding DNA sequence.

Although theoretically a genomic reference sequence seems best, in practice a coding DNA reference sequence is preferred.Good for promoters, alternative splicing etc.

When the entire genomic sequence is not known, a cDNA reference sequence should be used. To avoid confusion in the description of a variant it should be preceded by a letter indicating the type of reference sequence used.

Several different reference sequences can be used.

"C." for a coding DNA sequence (like c.76A>T) "g." for a genomic sequence (like g.476A>T)

"m." for a mitochondrial sequence (like m.8993T>C,)

"r." for an RNA sequence (like r.76a>u)

"p." for a protein sequence (like p.Lys76Asn)

Describing genes / proteins, only official HGNC (HUGO gene nomenclature committee) gene symbols should be used (www.genenames.org)

DNA reference sequence used should preferably be from the RefSeq database, listing both database accession and version number (like NM_004006.2)

There is no nucleotide 0 (zero).

Nucleotide 1 is A of the ATG-translation initiation codon.

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The nucleotide 5' of the ATG-translation initiation codon is -1, the previous -2, etc. e.g. in 5' UTR C.-12G>A

The nucleotide 3' of the translation stop codon is *1, the next *2, etc. e.g. in 3' UTR c.*70T>A

143. Define Genetic Disorders and its classification.(REPEAT)

Genetic Disorders: A genetic disease or disorder appears as a result of mutations in an individual's DNA. Change in DNA sequence may be small or large like deletions/addition of entire chromosome.

Fully penetrant and low penetrant

Fully penetrant, where other genes and environmental factors have no effect

Low-penetrance, genes play a small part, along with other environmental factors

Classification of genetic disorders

1) Multifactorial disorders

2) Single gene disorders

3) Chromosomal disorders

4) Mitochondrial disorders

5) Somatic mutations (cancers)

144. Explain Inheritance patterns of genetic disorders

Autosomal Dominant: Affected individuals are heterozygote. One allele is mutated, while other one is normal.

Autosomal recessive: Homozygous, when both alleles of the gene are mutated.

X-linked recessive: Mutated alleles are present on either of the sex chromosomes.

Multifactorial: Genes and environment both contribute. Multifactorial frequent, single gene disorders less frequent: Multifactorial are common. Single gene less common like dominant/recessive pedigree patterns

Chromosomal disorders:- Thousands of genes may be involved. Multiple organ systems affected at multiple stages in gestation

145. Explain Autosomal recessive inheritance / diseases

Autosomal Recessive diseases: Diseases occurs in individuals with two mutant alleles. In general, an individual inherit one mutant allele from each of the parent.

Recessive diseases: Homozygotes with two copies of the altered gene

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Recessive diseases: Since each parent has two alleles. Chance of inheriting a mutant allele from one parent is 50% and for the other parent is also 50%. Net chance of inheritance of two mutant alleles is 25%.

Generally, the disease appears in the progeny of unaffected parents.

Affected progeny include both males and females equally.

Phenotypic proportions are equal - autosomal recessive: When we know that both male and female phenotypic proportions are equal, we can assume that we are dealing with autosomal inheritance.

146. Write name of Common recessive disorders

Nucleotide numbering is purely arbitrary and starts with 1 at the first nucleotide of the database reference file.

No +, - or other signs are used.

The sequence should include all nucleotides covering the sequence (gene) of interest and should start well 5' of the promoter of a gene.

1) Bloom syndrome

2) Carpenter syndrome

3) Cystic fibrosis

4) Thalassemia

5) Many forms of mental retardation

6) Gaucher's disease

7) Glycogen storage diseases

8) Rotor syndrome

9) Many of eye diseases

147. Properties of Autosomal recessive inheritance

Consanguinity: These diseases appear where parents have common ancestors.

Same allele increases chances of inheriting disorders: The chance of inheriting identical alleles by an individual from both parents increases the chance of inheriting a recessive disorder.

Common genetic background: Small populations of individuals with a common genetic background may have increased risk of recessive diseases. Ashkenazi: Gaucher disease. Phenotype found more likely in siblings of proband than in other relatives. Males and females are equally affected. In most of the cases, parents of affected individuals are asymptomatic. Risk for each of the sibling being affected. Risk of disease for each sib is 25 %.

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148. Explain Autosomal dominant inheritance/ disorders

Autosomal dominant disorders: In autosomal dominant disorders, the normal allele is recessive and the abnormal allele is dominant. In a typical pedigree, every affected person has one affected parent. Equal numbers of affected females/males are expected. Male to male transmission is possible. One-half of the children of an affected individual are expected to have inherited the dominant allele.

149. List autosomal Dominant disorders

Pedigree analysis, main clues for identifying a dominant disorder is that the phenotype tends to appear in every generation of the pedigree. Affected fathers/mothers transmit the phenotype to both sons and daughters. Trait is common in the pedigree. Trait is found in every generation.

Autosomal Dominant diseases

1) Marfan syndrome
2) Huntington's disease
3) Retinoblastoma
4) Waardenburg syndrome
5) Myotonic dystrophy
6) Polycystic kidney disease
7) Achondroplasia
8) Polydactyly
9) Heredity hearing loss

150. Explain Dominance and Recessiveness? Define Hemizygous

Dominance and recessiveness are the properties of characters/traits, not genes. A character is dominant if it is expressed in the heterozygote and recessive if not.

Recessiveness: Cystic fibrosis is recessive because only homozygotes manifest it, whereas heterozygotes show the normal phenotype.

Hemizygous: Males are hemizygous for loci on the X and Y chromosomes, where they have only a single.

151. Explain X-linked recessive inheritance and disorders? Their Characteristics and disease. (REPEAT)

X-linked recessive: Traits which are due to the genes which are present on either of the sex chromosomes. Many of the genes have a disease phenotype.

Males are hemizygous and express the disease: Males are hemizygous, will express the disease phenotype if one mutation is present.

Homozygous Females manifest disease: Females may be homozygous or heterozygous. Homozygous may manifest the disease.

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General characteristic

1) Trait is rare in pedigree.

2) Trait skips generations. 3)Affected fathers do not pass to their sons.

4) Males are more often affected than females.

X-linked recessive disorders:-X-linked recessive: X-linked recessives expressed in all males but only in homozygous females.

Common X-linked disorders

1) Duchene Muscular Dystrophy
2) Hunter's Disease
3) Menkes Disease
4) Hemophilia A and B
5) Color Blindness

152. Write name of Common X-linked dominant diseases

1) Rett syndrome
2) Alport syndrome
3) Goltz syndrome
4) Fragile-X Syndrome

153. What is difference between mtDNA and Nuclear DNA ?

Nuclear DNA is inherited from all ancestors while mtDNA is inherited from a single lineage i.e from mother only.

154. What is Multifactorial inheritance and disorders?

Multifactorial diseases: Genetic disorders controlled by many genes plus the effects of the environment.

155. What is difference between genetic and environmental disorders in Multifactorial?

They are rare, genetically simple, unifactorial and high recurrence rate while environmental they are common genetically very complex, multifactorial and low recurrence rate.

156. writes the name of Multifactorial diseases (REPEAT)

1) Asthma
2) Cancers
3) Ciliopathies
4) Cleft palate
5) Diabetes
5) Heart diseases

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6) hypertension
7) inflammatory bowel disease
8) intellectual disability
9) mood disorder
10) Obesity
11) infertility

157. Explain Diagnostic tests for genetic disorders

Physical analysis: Physical analysis for affected individual.

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities.

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus

158. Define pedigree and its use.

A complete, detailed family history is called a pedigree. Pedigrees are used to determine the pattern of inheritance of a genetic disease within a family.

159. Define Gene therapy

Gene therapy a procedure that involves identification, manipulation, and transference of genetic segments into a host to replace defective genes and to perform desired genetic activities.

160. Explain Complications in basic inheritance pattern

Single genes -Non-mendelian inheritance

Differences in gene actions can generate more complex inheritance patterns.

Single gene can give codominance, incomplete dominance, overdominance.

Complications to inheritance patterns

Multiple genes - epistasis, polygenic traits Genes & the environment - sex-influenced traits, incomplete penetrance

Complications

1) A common recessive inheritance can give dominant pattern 2) Autosomal dominant inheritance with non-penetrance 3) Autosomal dominant inheritance with variable expression.

161. What are the Goals/Reasons of pedigree analysis?

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1) Determine the mode of inheritance: dominant, recessive, sex-linked.

2) Determine the probability of an affected offspring for a given cross

162. Define DNA polymorphism and its phenotypic effects.

DNA polymorphism is a sequence difference compared to a reference standard present in at least 1–2% of a population. DNA Polymorphism may be single nucleotide or thousands of BASES. DNA Polymorphism may or may not have phenotypic effects

163. Polymorphic sequences can serve as markers? Justify

If the location of a polymorphic sequence is known, it can serve as a marker for locating other genes or genetic regions. Each polymorphic marker has different versions or alleles. Polymorphisms can be silent, or be exhibited at levels of Proteins. linked SNPs outside of gene :- have no effect on protein production or function.

Causative SNPs in gene:

Non coding SNP:change amount of protein produced.

Coding SNP change amino acid sequence

164. Why to study- Polymorphism

Essential to study inheritance patterns. Map phenotypes and anchor genes to the genetic map by co-segregation analysis. Determine change in function: resistant/sensitive populations genetically determined variability among humans is due to a difference in 0.1% of the genomic sequence.

165. what are the Types of Polymorphic DNA Sequences

RFLP: Restriction Fragment Length Polymorphism

VNTR: Variable Number Tandem Repeats (8 to > 50 base pairs)

STR: Short Tandem Repeats (1-8 base pairs).

SNP: Single Nucleotide Polymorphisms

AFLP: Amplified Fragment Length Polymorphism

166. write the Applications of molecular markers (REPEAT)

– short range

1) Parentage determination 2) Genetic distance estimation 3) Determination of twin zygosity implanted embryos 4) Sexing of pre 5) Identification of disease carriers

- Long Range

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1) Gene mapping & mapping of QTL by linkage 2) Markers assisted selection of breeds

167. Explain Restriction Fragment Length Polymorphism and their uses in detailed

What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples.

Most RFLP markers are heterozygous and highly locus-specific.

RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis.

Methodology

Genomic DNA digested with restriction enzymes. DNA fragments separated via electrophoresis and transfer to nylon membrane. Membranes exposed to probes labeled with P32. Film exposed to X-Ray

Uses

RFLPs can be used in paternity cases or criminal cases to determine the source of a DNA sample. (i.e. it has forensic applications). RFLPs can be used to measure recombination rates which can lead to a genetic map with the distance between RFLP loci. Direct method for selecting desirable genes such as disease resistance genes RFLP used to determine disease status • Determine disease status of an individual, i.e.Huntington • Cystic fibrosis • Sickle cell anemia Mutation detection • RFLPs can be used to determine the disease status of an individual; (e.g. it can be used in the detection of particular mutations)

168. Define VNTR (REPEAT)

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes Show variations in length between individuals. Each variant acts as an inherited allele, allowing them to be used for personal or parental identification.

169. Write the Importance of VNTR

VNTRs are an important source of RFLP genetic markers used in linkage analysis (mapping) of genomes. When removed from surrounding DNA by the PCR or RFLP methods, and their size determined by gel electrophoresis or Southern blotting. As a result, they produce a pattern of bands unique to each individual.

170. Write uses of Molecular Markers

1) Different types of markers used to study diseases

2) DNA forensics

3) QTL traits in animals and plants

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171. Explain SNP-Screening and its importance

People can be grouped based on SNP profile

1) Genome of each individual contains distinct SNP pattern.

2) People can be grouped based on the SNP profile.

SNP - found in coding and noncoding regions

1) SNPs are found in coding and (mostly) noncoding regions.

2) Occur with a very high frequency about 1 in 1000 bases to 1 in 100 to 300 bases.

SNP SCREENING methods: Two different screening strategies

1) Many SNPs in a few individuals

2) A few SNPs in many individuals

SNP Importance

1) In disease diagnosis 2) In finding predisposition to diseases

3) In drug discovery and development

4) In drug responses

5) Investigation of migration patterns

6) Medication and diagnosis at individual level

172. What is the Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 -9.5 per 1000 individuals

173. What is the Location of intronic sequence with respect to exons

Beginning of intron

The number of the last nucleotide of the preceding exon, a plus sign and the position in the intron, like c.88+2T>G

End of intron

The number of the first nucleotide of the following exon, a minus sign and the position upstream in the intron, like C.89-1G>T

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